AI Assisted Coding

Assignment 14.3

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Batch:45

Task-1:

Scenario

A student wants to showcase academic projects, technical skills, and contact information through a personal portfolio website. AI tools are used to speed up development while maintaining quality and customization.

Tasks

1. Use GitHub Copilot to generate a semantic HTML structure with sections:

o Home

o About

o Projects

o Contact

2. Ask Copilot to suggest responsive CSS using Flexbox or Grid.

3. Customize AI-generated CSS to add:

o Hover effects on project cards

o Smooth scrolling navigation

Expected Outcome

• Clean and well-structured portfolio website

• Responsive layout for desktop and mobile screens

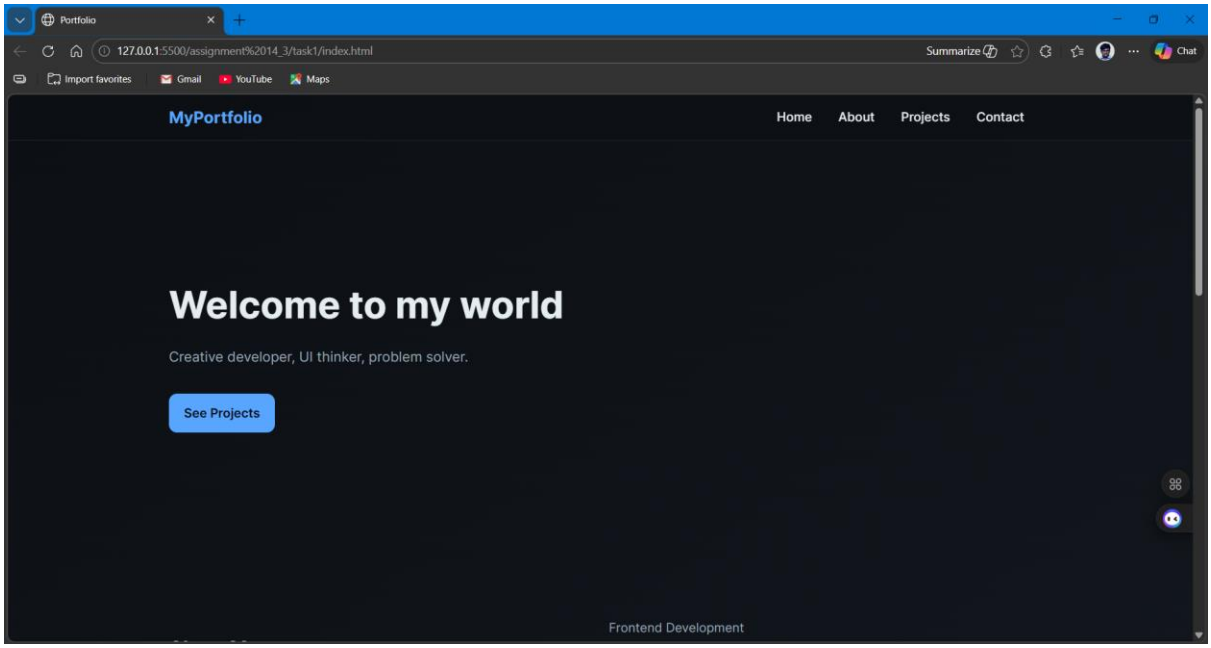
• Interactive project cards

Output:

index.html U

Assignment_14-3 > task1 > index.html > ...

```html
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <link rel="icon" href="https://www.cloudflare.com/favicon.ico" />
    <title>Not Found</title>
    <style>
      body {
        font-family: system-ui;
        font-weight: 300;
        font-size: 1.25rem;
        color: #36393a;
        display: flex;
        align-items: center;
        justify-content: center;
      }
      main {
        max-width: 1200px;
        margin-top: 120px;
        display: flex;
        flex-wrap: wrap;
        align-items: center;
        justify-content: center;
      }
      #text {
        max-width: 60%;
        margin-left: 1rem;
        margin-right: 1rem;
      }
      main > section > div {
        margin-bottom: 3.25rem;
      }
```



Task-2:

Scenario

A local restaurant requires a simple yet attractive landing page to display its offerings. The developer uses AI assistance to rapidly generate UI components.

Tasks

1. Use Copilot to create a navigation bar with links:

o Home

o Menu

o Gallery

o Contact

2. Generate a menu section styled with CSS cards.

3. Implement a JavaScript-based image slider for the gallery using Copilot suggestions.

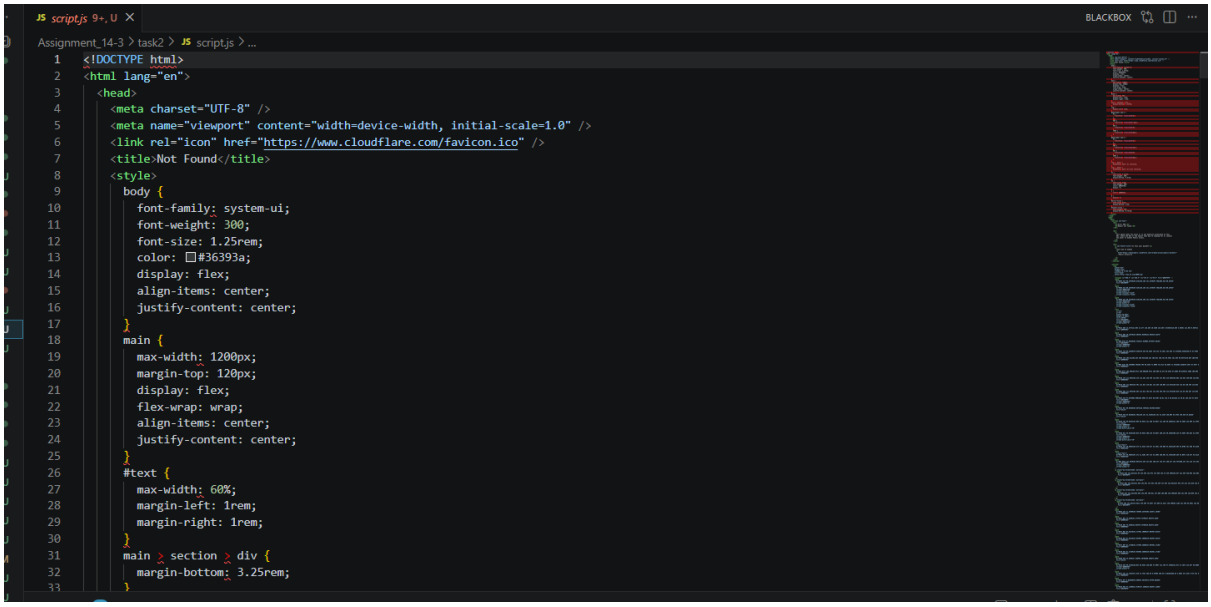
4. Customize animations or transitions for better user experience.

Expected Outcome

• Fully functional restaurant landing page

• Responsive navigation bar

• Working image slider with smooth transitions



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<!DOCTYPE html>
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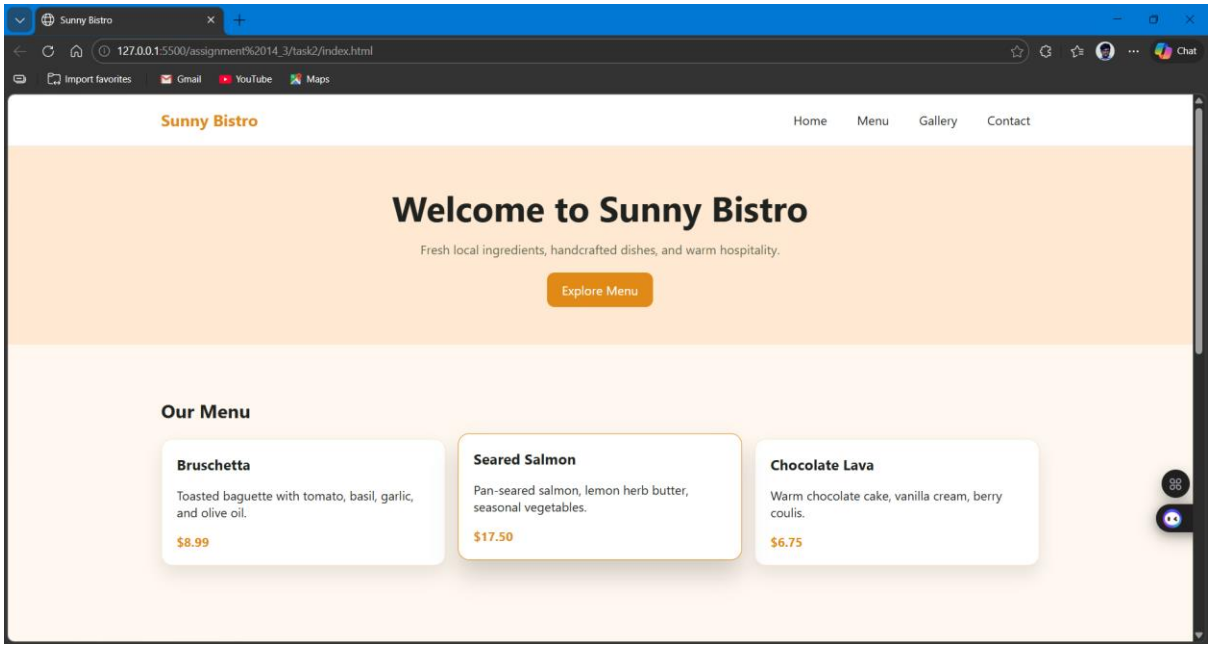


Task 3: AI-Powered Event Registration Form

Scenario

SR University is hosting a technical fest and requires an online registration form with real-time validation to ensure accurate user input.

Tasks

1. Ask Copilot to generate an HTML form with the following fields:

o Name

o Email

o Phone Number

o Department

o Event Selection

2. Use AI assistance to style the form using CSS for readability and accessibility.

3. Implement JavaScript validation using Copilot:

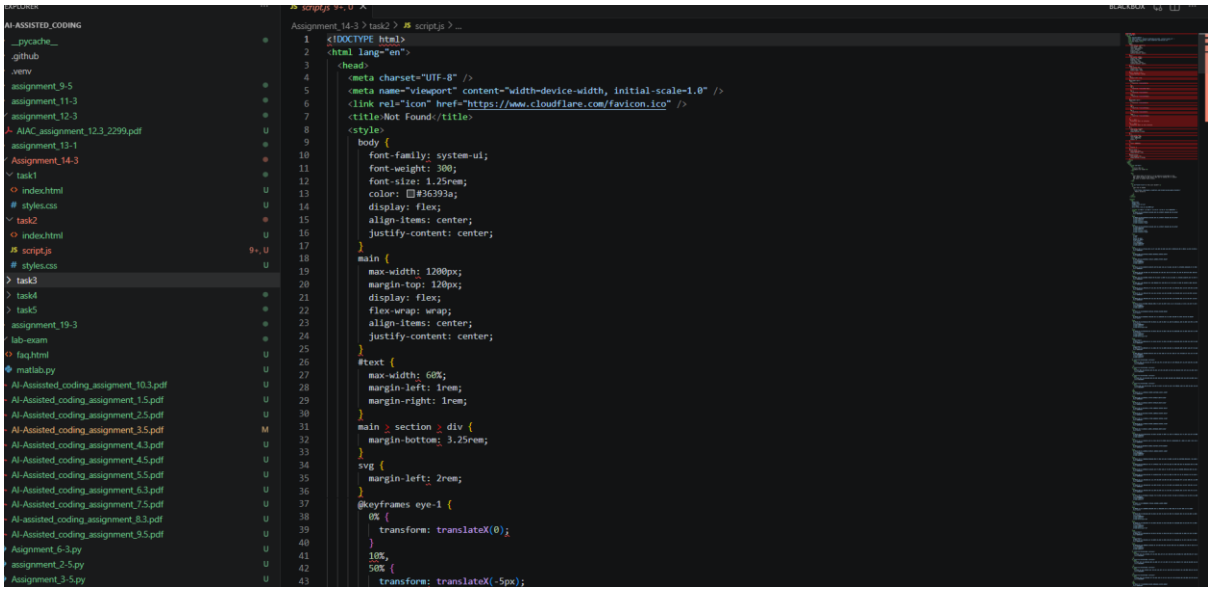
o Email format validation

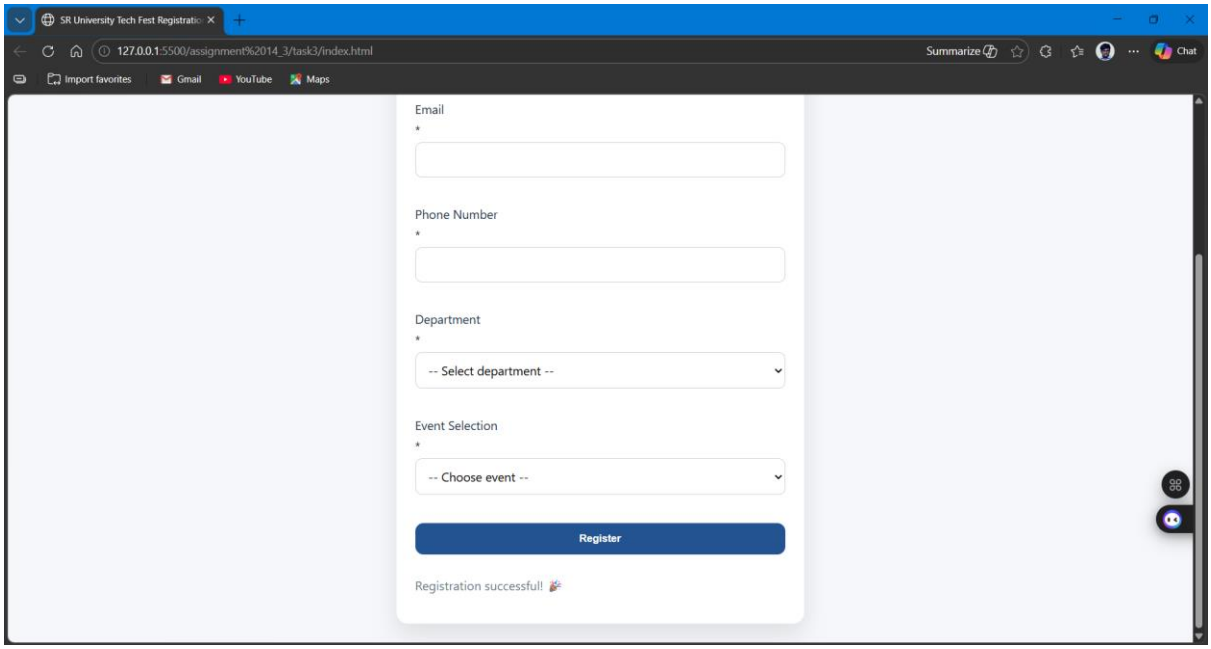
o Phone number length validation

o Required field checks

Expected Outcome

- User-friendly and visually appealing form
- Real-time validation messages
- Prevention of invalid submissions





Task 4: AI-Assisted E-Commerce Product Page

Scenario

A startup is developing a basic e-commerce product page to showcase products and manage simple cart functionality.

Tasks

1. Use Copilot to generate a grid-based product catalog using HTML and CSS.

2. Implement an “Add to Cart” feature using JavaScript.

3. Modify AI-generated code to include:

o Cart item counter at the top-right corner

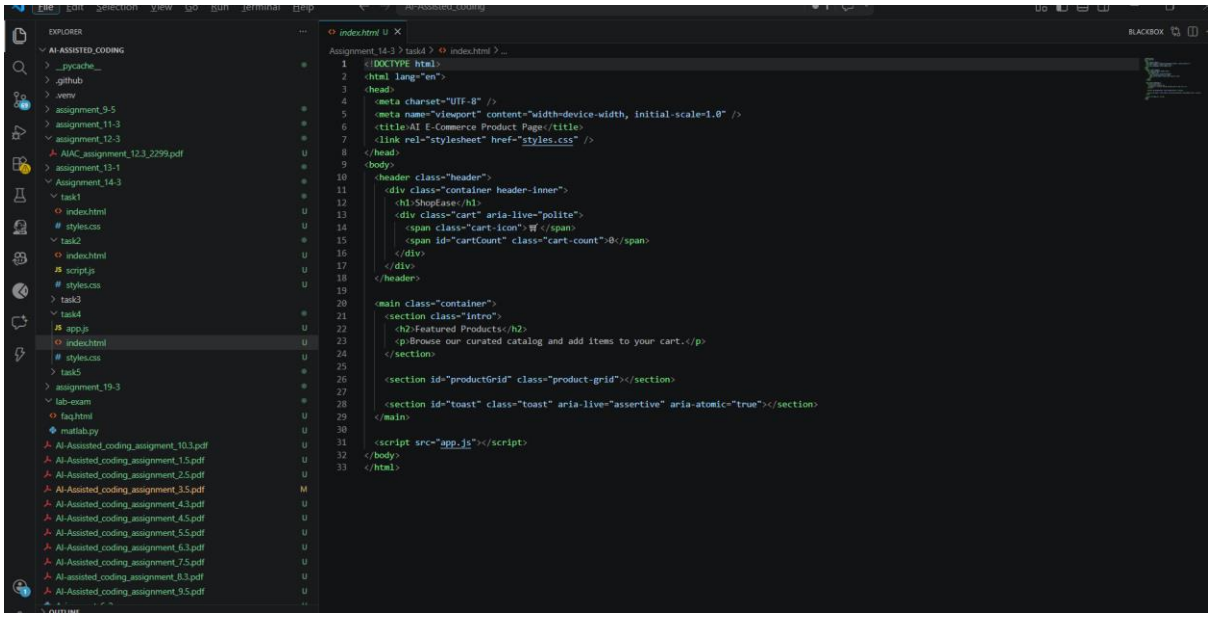
o Visual feedback when a product is added

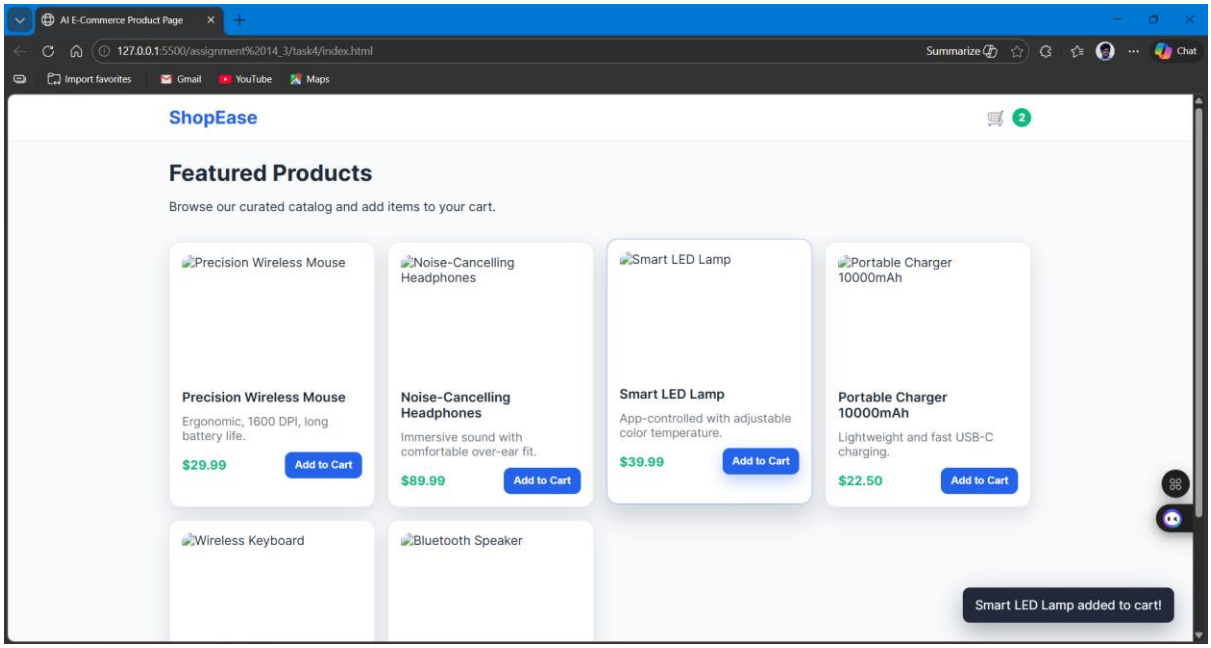
Expected Outcome

• Functional product catalog

• Working cart counter

• Interactive user experience





Task 5: Create a Responsive Web Page Layout

Instructions:

• Design a basic web page layout with a header, main content area, and footer using HTML and CSS.

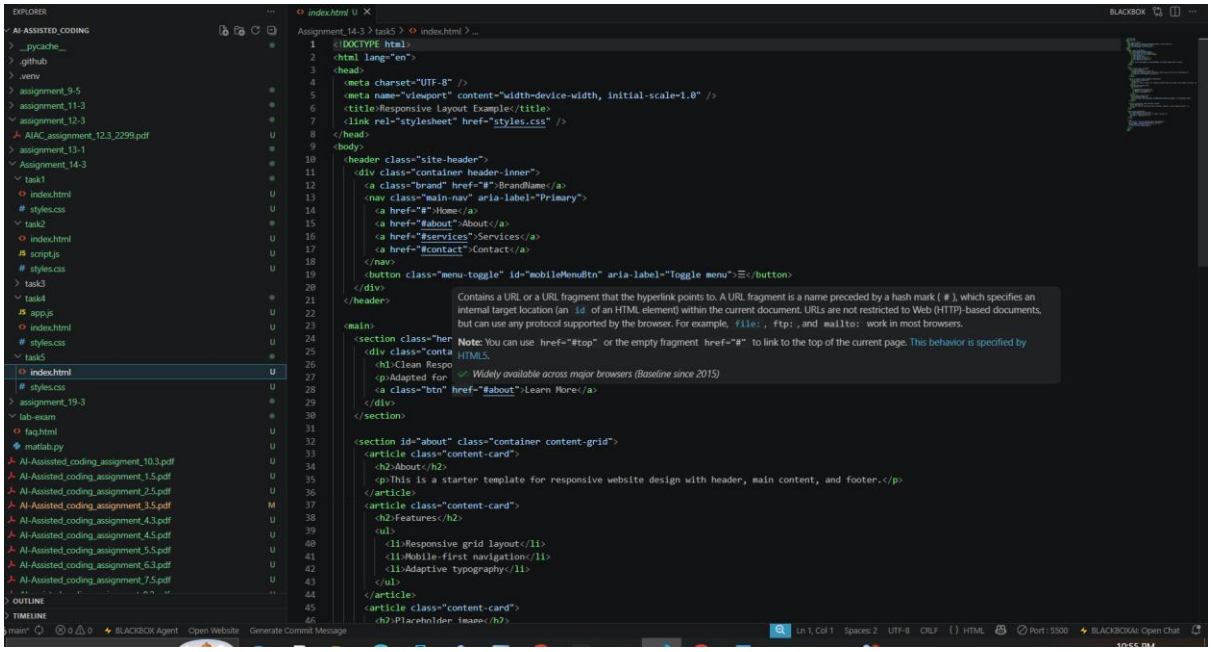
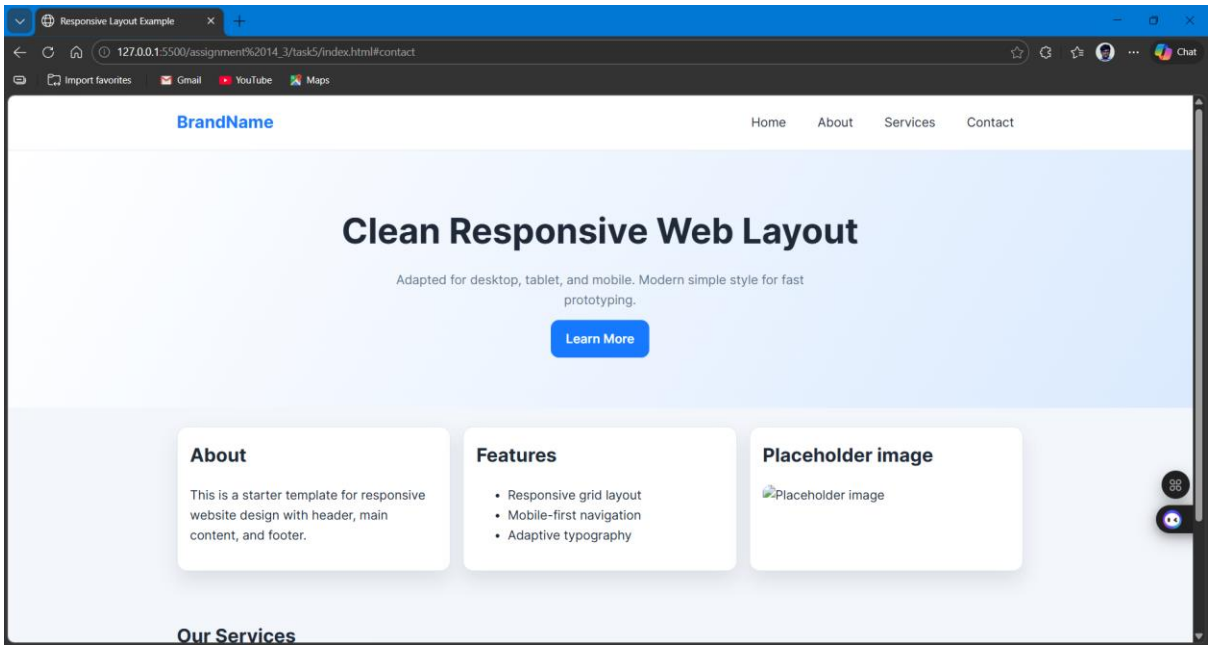
• Use AI to assist in generating responsive CSS for different screen sizes.

• Ensure the layout is clean and visually organized.

Expected Output:

• A responsive web page with:

• Header with navigation links

• Main section with placeholder text/images

- Footer with copyright or contact info
- Layout adapts correctly to desktop, tablet, and mobile screen sizes.